

Hybridisation is another key enabler of today's printed and flexible electronics

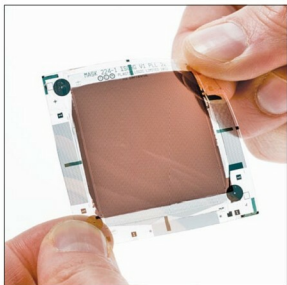
Hybridisation is a key word in the printed electronics space right now. It is the combination of traditional electronics technologies and components with newer printed electronics technologies. Innovative companies such as American Semiconductor in Boise, Idaho, are offering new flexible silicon based processors for use in wearables and other applications.

PragmatiC in the UK continues to move ahead with organic based processors, and Thin Film Electronics based in Norway is regularly in the news for applications utilising their printed memory.

Each of these companies is working with methods to attach their products into more complex circuit designs, utilising both existing and new connection technologies such as solder or conducting adhesives.

Ohmtext in Denmark has spent extensive time developing successful means to connect electronics with textiles for applications ranging from sports and athletic use to industrial and first-responder safety gear. Products for these groups range from various forms of wearable devices to new applications of near-field-communication/sensor combinations.

—Stan Farnsworth, VP - marketing, NovaCentrix



Plastic Logic and ISORG's sensor which promises to transform the way we interact with consumer devices (Image courtesy: www.isorg.fr)

Sensor that could revolutionise industrial and consumer electronics.

The fully-flexible approach targets a more unobtrusive and large-area integration of developed systems. A great example of this latter approach includes the multi-awarded image sensor on plastic, unveiled by ISORG and Plastic Logic, which combines the flexible large-area photodetector technology from ISORG and the organic thin-film technology from Plastic Logic.

This opens the way to new applications, ranging from smart packaging, sensor tags for medical and biomedical applications, security

and mobility commerce, to environmental and industrial electronics.

Fascinating research and development happening

One of the most challenging goals for flexible electronics is to achieve lightweight and unobtrusive devices, notes Giuseppe Cantarella, PhD student, Wearable Computing Laboratory, Swiss Federal Institute of Technology, Zurich. Keeping high electrical performances in mind, researchers are now taking big steps forward to

integrate with respect to customers' attitudes and requirements.

He says, "The new trends now go in specific directions to fulfil the market focus. One of these is bio-compatibility for implantable devices that can remotely monitor and improve our healthcare, provide mechanical and electrical stability over time and low cost for the development of a technology that is accessible to most."

Materials and manufacturing techniques are the two main areas for research. The main focus of research activities in universities and research centres in the field of print-

able and flexible electronics is in two main directions: the development of new materials with better performances and low-cost deposition, and the study of new technologies and manufacturing techniques on plastic substrates with superior mechanical properties.

"Since next-generation electronics is expected to be conformal to any surface including human skin or human tissues, many efforts have been devoted to establish engineered techniques for conformal electronics," Cantarella says. A great example of this approach has been demonstrated by the group of Prof. Someya of Tokyo University, where implantable bio-compatible devices have been realised using organic semiconductors.

In parallel to the development of graphene, nanotubes and 2D materials, new material formulations have been introduced. A very new and interesting work is the one on printed liquid silicon on paper, demonstrated by the Delft team, led by Prof. Ishihara at Delft University of Technology. These results could lead to high-performance and low-cost printed and flexible transistors and circuits.

Smart contact lens to inform diabetic patients on blood-glucose levels. A brilliant example of tackling a growing problem of diabetes using flexible and printed electronics is Google's Smart Contact Lens project, co-developed by Google and Novartis.

Uncontrolled blood sugar poses a threat to people's lives and could damage their eyes, kidneys or heart in the long run. People with diabetes, more often than not, have fluctuating glucose levels. Sudden spikes or drops are dangerous. To monitor the blood-glucose level, people with diabetes need to prick their finger and test drops of blood at regular intervals throughout the day.

In hopes of finding easier methods to measure glucose body levels, scientists have been researching on an alternate, non-invasive way in